Nermine Moumen

Industrial Chemical Engineering Student

• *nermine.moumen@insat.ucar.tn* • +21627040414 • *linkedin.com/in/nermine − moumen*

EDUCATION

National Institute of Applied Sciences and Technology (INSAT) *2022–Present*

- Final-year Industrial Chemical Engineering student, specialized in Process and Environment, with a foundation in Polymer Chemistry, Materials Science, and Organic Chemistry, complemented by Analytical Chemistry, Physicochemical Characterization, and Electrochemistry. Seeking a final-year internship (PFE).

PROFESSIONAL EXPERIENCE

SEGULA Technologies – Electrochemical Battery Modeling Internship *06/2026–08/2026*

- Modeled a **Li-ion cell** (NMC811/Graphite-SiOx) using the Single Particle Model (SPM) implemented in MATLAB.
- Solved the governing PDEs using the **Finite Volume Method** (FVM) with Crank–Nicolson time integration.
- Optimized the mesh and diffusion coefficient values, reducing the RMSE error **from 202.13 mV to 40.5 mV**.
- Extracted and processed **experimental discharge data** from a database to validate MATLAB and COMSOL results.

Control Union – Quality Control Internship *08/2025–09/2025*

- Analyzed lubricants for viscosity and heavy metals **(ICP-OES)**.
- Prepared and interpreted **analytical reports** to verify compliance with ASTM D5185 and **ISO 17025** standards.

ETAP – Water Treatment and Petrochemistry Internship *07/2024–08/2024*

- Performed physico-chemical and ionic analyses of water samples (pH, salinity, chlorides via ion chromatography).
- Studied water treatment methods **(coagulation-flocculation, ion exchange)** for contaminant and corrosion control.
- Performed organic geochemistry techniques **(GC, GC-MS, Rock-Eval pyrolysis)** for hydrocarbon **characterization**.

ADDITIONAL EXPERIENCE

Cascade – Online Chemistry Teacher (Part-time) *2024–Present*

- Provide individualized online chemistry tutoring for French Terminale students, covering exam-level concepts.

ACADEMIC PROJECTS

Molecularly Imprinted Polymers (MIPs) for Pharmaceutical Applications

- Synthesis and characterization of a sulpiride-selective MIP via radical polymerization, with Soxhlet extraction, UV-Vis spectrophotometry monitoring, and validation through controlled-release kinetics (240 min)
- Precipitation synthesis of a spherical MIP selective to ribavirin, validated on human bronchial cells
- Synthesis of a core-shell MOF/polymer MIP for targeted doxorubicin delivery, achieving 85% encapsulation efficiency

Superconducting Materials: From Synthesis Routes to Structural Characterization

- Analyzed the fundamental properties of superconductors and studied elaboration routes including solid-state ceramic synthesis, crystal growth (flux, Bridgman, Czochralski), and **thin-film deposition (ALD)**.
- Reviewed characterization methods: **RAMAN, FTIR, XRD, SEM and TEM.**

Prediction of Liquid–Liquid Equilibria and Cross-flow and Countercurrent Extraction for Ternary Systems

- Formulated the LLE thermodynamic system and solved it via Rachford-Rice and **arc-length continuation**.
- Reproduced 7 Mathematica notebooks proving equivalence of the **Hunter-Nash and McCabe-Thiele** methods.

TECHNICAL SKILLS

- **Lab Techniques:** Chromatography (HPLC, GC, TLC), Spectrometry (MS, ICP-OES), Spectroscopy (NMR, FTIR, UV-Vis), Diffraction (XRD), Microscopy (SEM, TEM), Electrochemistry (Cyclic voltammetry, Polarography).
- **Simulation:** Aspen Plus, Aspen HYSYS, COMSOL Multiphysics • **Programming:** $MATLAB, Mathematica, Python$
- **Data & Office:** Pack Microsoft Office, OpenLCA • **Standards:** $ISO9001, ISO14001, ISO45001, ISO17025, ISO14040$

EXTRACURRICULAR ACTIVITIES

- **IEEE INSAT** – Chair of EMBS; Organization team leader of TRSYP 3.0; Social activities manger of MENASYP
- **THEATRO INSAT** – Member of the dancing team

LANGUAGES

English (Fluent) French (Fluent) Arabic (Native)

CERTIFICATES

Aspen HYSYS – MATLAB